

RAHUL PODDAR

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SUMMARY

Senior Backend Software Engineer with 6+ years building data-intensive backend systems in Java and Spring Boot. Proven track record delivering REST APIs, high-volume data ingestion pipelines, RBAC authorization, and database performance optimizations in production healthcare platforms. Hands-on experience designing Kafka-based event-driven microservices through distributed systems projects.

TECHNICAL SKILLS

Languages & Databases:	Java, Oracle (SQL, PL/SQL), PostgreSQL, MySQL
Backend & APIs:	Spring Boot, Spring MVC, JPA/Hibernate, MyBatis, RESTful APIs
Architecture & Patterns:	Distributed Systems, Event-Driven Architecture, Caching (Caffeine, Redis), Role-Based Access Control
Servers & Build:	Kafka, Docker, Maven, Git, Apache Tomcat, Oracle WebLogic, Zipkin (Distributed Tracing)
Testing & Quality:	JUnit, Mockito, SonarQube, JaCoCo
Frontend:	Thymeleaf, JavaScript, jQuery, AngularJS, HTML, CSS, AG Grid

WORK EXPERIENCE

Smart Source Technologies Inc

Albany, NY

Senior Software Engineer

Feb 2021-Present

Project- IMAPS (Information Management Application for Partner Services)

- Owned backend development of IMAPS, a centralized assignment and investigation platform modernizing a legacy public-health MIS system, used for managing workflows across multiple disease programs (HIV, Syphilis, Chlamydia, Gonorrhea, Mpox).
- Designed Spring Boot REST APIs and service-layer workflows, integrating complex Oracle PL/SQL stored procedures, and optimizing queries for high-volume investigation workflows, while enforcing secure access control using Spring Security RBAC.
- Improved backend scalability and reduced database load by introducing caching (Spring Cache/Caffeine) for frequently accessed data.
- Diagnosed investigation workflow latency using Zipkin-based tracing with correlation IDs and optimized it by eliminating redundant database writes, adding proper indexing, and implementing pagination — reducing endpoint latency from ~500ms to ~20–40ms.
- Built unit tests with JUnit and Mockito, mocking dependencies to validate business logic and improve code quality, supported by SonarQube coverage tracking.

Project- NYEHMS (New York Electronic HIV Monitoring System) Web Application (Continued support)

- Redesigned data ingestion workflows to support SFTP-based file processing alongside HL7/HIE feeds, ensuring uninterrupted ingestion during a production transition.
- Investigated production ingestion failures and bottlenecks — redesigned a long-running HL7 deduplication job from per-record to batched queries and built tooling (job runners, ingestion alerts) to improve reliability and observability.

Health Research Incorporated

Albany, NY

Application Development Specialist III

May 2019-Feb 2021

Project- NYEHMS (New York Electronic HIV Monitoring System) Web Application

- Contributed to backend development of NYEHMS, a production statewide HIV monitoring platform ingesting high-volume clinical data feeds (CCD XML, HL7 messages, and death records) from regional healthcare partners.
- Built data ingestion workflows to parse HL7, CCD XML, CSV, and text-based healthcare documents, processing thousands of records daily and transforming them into normalized patient and laboratory records using Java and Oracle PL/SQL.
- Implemented parsing and normalization logic using JDOM (XML) and HAPI HL7 libraries to extract patient demographics, encounters, diagnoses, and laboratory results into structured database tables.
- Designed error-recovery workflows using HIT_ERROR tracking tables, enabling failed records to be reviewed, corrected, and safely reprocessed during nightly batch runs.

Software Developer/Engineer

May 2018-May 2019

Project – Environmental Public Health Tracking Application

- Developed backend business logic and RESTful services using Java, Spring Boot, and Hibernate for public-health applications.
- Integrated backend services with AngularJS-based UI components and supported enterprise deployments on Oracle WebLogic Server.

DISTRIBUTED SYSTEMS PROJECT

Event-Driven Case Management System (Kafka Microservices)

Technologies: Java, Spring Boot, Kafka (Redpanda), PostgreSQL, Docker, Flyway

- Designed an event-driven investigation platform using Kafka to model patient resolution, case lifecycle, and lab ingestion workflows.
- Designed loosely coupled microservices using Kafka topics, partitioning strategies, and schema contracts to ensure ordering and scalability.
- Implemented idempotent consumer processing and transactional guarantees ensuring correctness under Kafka's at-least-once delivery.
- Engineered **fault-tolerant ingestion pipelines** with retries, DLQ handling, and persistent error tracking for observability and replay.

EDUCATION

Executive Master of Business Administration, University of the Cumberlands

May 2021-Dec 2022

Master of Science (MS), Computer Science, University at Albany SUNY

Aug 2016-May 2018

Bachelor of Engineering, Computer Science, Madhav Institute of Technology & Science

Aug 2011-May 2015